MODULE 1 · LECTURE 24

AI-Based Autonomous Crop and Livestock Management

How autonomous machines sense, perceive, decide and act; how machine learning is trained and judged; how autonomous tractors, field robots, milking robots and animal monitors work — with a work-rate example, a heat-detection program for dairy cows and the evaluation metrics used throughout AI.

Learning objectives

After this lecture you should be able to:

1. Explain autonomy, the sense–perceive–decide–act loop and the levels of automation in farm machinery.
2. Describe the main types of machine learning, the steps of building a model and how models are evaluated.
3. Describe the components and uses of autonomous tractors and field robots.
4. Explain precision livestock farming, automatic milking and animal monitoring.
5. Calculate work rates of robot fleets and evaluation metrics (precision, recall, F1) for an AI alert system.
6. Discuss the benefits, risks, safety requirements and Indian context of autonomous farming.

Scope: Lecture 19 introduced AI and intelligent agents, Lectures 21–22 search and path planning, and Lecture 23 the IoT that feeds AI with data. This lecture explains how machines learn from that data and act on their own in crop and livestock production. Disease and pest detection (Lecture 25), yield prediction (Lecture 26), weeding and harvesting robots (Lecture 27) and sorting (Lecture 28) build on it.

1. From automation to autonomy

An **automated** machine repeats a programmed action — a timer opens a valve at 6 a.m. An **autonomous** machine decides for itself how to reach a goal in a changing environment: it senses the field or the animals, interprets what it senses, plans what to do and acts, again and again, with a person supervising rather than controlling every step. This is the intelligent-agent idea of Lecture 19, built into tractors, robots, feeders and milking stalls.

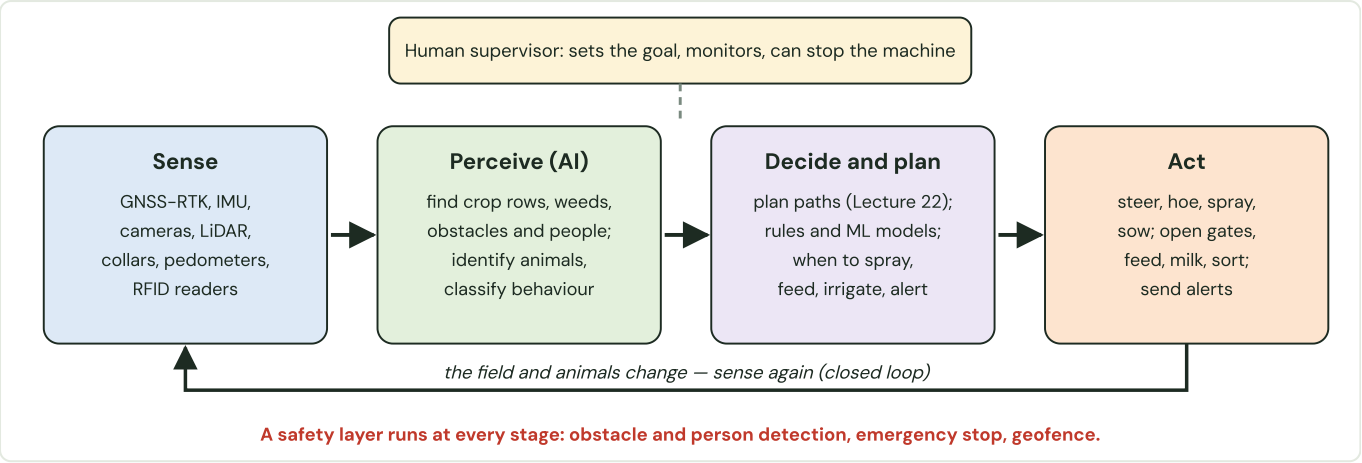


Figure 1. The sense–perceive–decide–act loop of an autonomous farm system, with crop and livestock examples. Original diagram.

Levels of automation

Level	What the machine does	Role of the person	Example
0. Manual	Nothing by itself	Does everything	Conventional tractor, hand milking
1. Assisted	Gives information or warnings	Drives and decides	GNSS guidance light-bar; heat-alert app
2. Partially automated	Controls some functions	On board, supervises, takes over at any time	Auto-steer with section control; automatic cluster removal in a milking parlour
3. Supervised autonomy	Performs the whole task	Watches nearby or remotely; can stop the machine	Driverless tractor working a field; robotic milking stall
4. Full autonomy	Plans and performs work, handles most exceptions	Sets goals; called only for problems	Fleets of small robots planning their own tasks (mostly research today)

This five-level scale is a teaching aid for this course, not a formal standard. Safety design of such machines is covered by ISO 18497-1:2024, which sets principles for agricultural machinery and tractors with partially automated, semi-autonomous and autonomous functions (International Organization for Standardization [ISO], 2024).

2. How machines learn: machine learning in brief

In an expert system (Lecture 17) people write the rules. In **machine learning (ML)** the computer finds the rules — the patterns that link inputs to outputs — from examples. By applying ML to sensor data, farm management systems are becoming real-time, AI-enabled programs that give recommendations for decisions and actions (Liakos et al., 2018).

Type of learning	How it learns	Farm examples
Supervised	From labelled examples: inputs with the correct answer	Classification: healthy or diseased leaf, heat or no heat. Regression: yield or milk from weather, soil and feed data
Unsupervised	Finds groups or structure in unlabelled data	Grouping fields into management zones; grouping cows by behaviour
Reinforcement	Learns actions by trial, guided by rewards and penalties	Learning an irrigation or robot-steering policy that saves water or keeps to the row (Russell & Norvig, 2020)

Liakos et al. (2018) classified research on ML in agriculture into crop management (61% of applications — mostly disease detection, 22%, and yield prediction, 20%), livestock management (19%), water management (10%) and soil management (10%). For images, **deep learning** — neural networks with many layers — has become the leading method: a survey of 40 studies found that it gave high accuracy and outperformed commonly used image-processing techniques (Kamilaris & Prenafeta-Boldú, 2018). Lecture 25 explains it further.

Building and judging a model

1. **Define the task**: what must be predicted, from which data, and what a mistake costs.
2. **Collect and label data**: e.g. images marked "weed" or "crop"; days marked "heat" from breeding records.
3. **Split the data**: a training set to learn from and a test set kept aside to judge the model on cases it has never seen.
4. **Train** the model and tune its settings.
5. **Evaluate** on the test set with suitable metrics.
6. **Deploy** it on the machine or app, then **monitor** it and retrain when conditions change — a new breed, crop variety or season.

For a yes/no decision, results are counted in a **confusion matrix**: true positives (TP, correct alerts), false positives (FP, false alarms), false negatives (FN, missed cases) and true negatives (TN). From these:

$$\text{Accuracy} = (\text{TP} + \text{TN}) \div \text{all cases} \qquad \text{Precision} = \text{TP} \div (\text{TP} + \text{FP}) \qquad \text{Recall} = \text{TP} \div (\text{TP} + \text{FN}) \qquad \text{F1} = 2\text{PR} \div (\text{P} + \text{R})$$

Precision asks "when the system raises an alert, how often is it right?"; **recall** asks "of all the real cases, how many did it find?". The F1 score combines the two.

3. Autonomous crop management

Autonomous tractors and field robots combine the loop of Figure 1 with accurate positioning. Real-time kinematic (RTK) correction of satellite positions gives a location accuracy of a few centimetres — up to 3 cm for one commercial fertilising robot described by Oliveira et al. (2021) — enough to drive along crop rows.

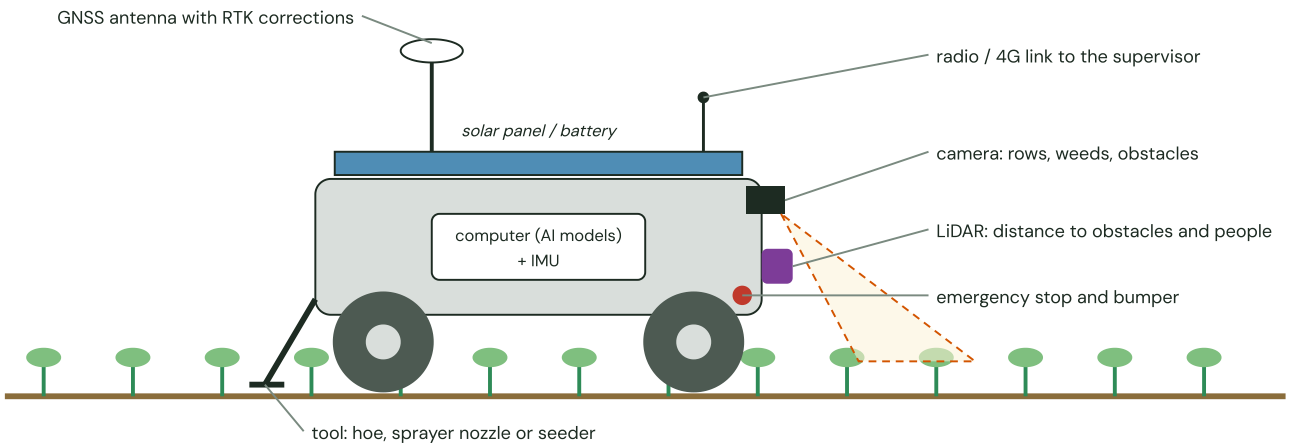


Figure 2. Main components of a small autonomous field robot. Original schematic.

Function	Technology	Role
Localisation	GNSS with RTK corrections, inertial measurement unit (IMU), wheel encoders	Knows where it is, even when satellite signals are briefly blocked
Perception	RGB and multispectral cameras, LiDAR, ultrasonic sensors, with AI models	Finds rows, crops, weeds, obstacles and people
Planning	Coverage path planning for whole fields; point-to-point planners such as A* (Lecture 22)	Decides the route and the order of tasks
Control	Steering, speed and implement controllers (closed loop, Lecture 12)	Keeps on the planned path and works the tool
Communication	Radio, 4G, LoRaWAN (Lecture 23)	Sends status and alerts to the supervisor; receives tasks
Safety	Bumpers, emergency stops, LiDAR protective zones, geofences	Stops before harming people, animals or property (ISO, 2024)

Oliveira et al. (2021) reviewed about 62 agricultural robotic systems for land preparation, sowing, planting, plant treatment, harvesting, yield estimation and phenotyping. Of these, 80.65% were still in the research stage, 22.06% were used for weeding and 64.52% had no robotic arm; the authors identified locomotion, sensors, computer-vision algorithms and communication as the main areas needing further work, and noted that financial cost is the biggest barrier for small producers.

Other autonomous crop systems

- **Closed-loop irrigation and fertigation**: sensor networks and weather forecasts drive valves and dosing pumps (Lecture 23).
- **Drones** that fly planned routes to map or spray fields (Lectures 15 and 25).
- **Greenhouse climate control** that adjusts vents, fans, screens and heating automatically (Lecture 12).
- **Robots for weeding, harvesting and sorting** (Lectures 27 and 28).

Worked example 1 — can small robots match a tractor?
Effective field capacity (ha/h) = speed (km/h) × working width (m) × field efficiency ÷ 10.
Tractor and 3 m seed drill: 6 km/h, efficiency 0.70 → 6 × 3 × 0.70 ÷ 10 = 1.26 ha/h; with a driver for 8 h a day → **10.1 ha/day**.
Small autonomous robot, 1.5 m wide: 4 km/h, efficiency 0.80 → 4 × 1.5 × 0.80 ÷ 10 = 0.48 ha/h; working 20 h a day, with breaks for recharging → **9.6 ha/day**.
Three robots supervised by one person → 3 × 9.6 = **28.8 ha/day**: a 60 ha sowing job takes about 2 days instead of 6.
Autonomy removes the limit set by the driver's working hours, and light machines put less load on the soil; against this stand purchase cost, supervision, repairs and the need for reliable positioning.

Speeds, widths, efficiencies and working hours are illustrative assumptions. A review of economic studies found that all 18 studies examined showed some scenarios in which automation and robotics were profitable, but most used partial budgets and prototype data, and all were set in developed countries; research for developing-country farming is still needed (Lowenberg-DeBoer et al., 2020).

4. Autonomous livestock management

Precision livestock farming (PLF) aims at real-time monitoring and management that warns the farmer when a problem arises so that immediate action can be taken, with continuous, automatic monitoring of animal health and welfare, product yield and environmental impact (Berckmans, 2014). ML is applied to animal welfare and livestock production (Liakos et al., 2018), and sensors, big data and ML can help farmers lower costs, raise efficiency and improve welfare (Neethirajan, 2020).

System	What it senses or does	Autonomous decision
Animal identification	RFID or barcoded ear tags read automatically	Links each animal to its own records, ration and alerts
Wearable sensors	Activity, rumination, temperature from collars, leg tags or boluses; also sound and image analysis (Neethirajan, 2017)	Alerts for heat, illness or lameness
Automatic milking system (AMS)	Cow enters the stall when she chooses; robot attaches cups; milk yield and quality recorded	Decides whether a cow is due for milking; flags abnormal milk
Automatic feeding	Feed mixed, delivered and pushed up by machines; RFID feeders (Lecture 12)	Gives each animal its ration by yield or stage
Camera monitoring	Body condition, gait, behaviour in the shed	Early warning of lameness, injury or distress

Automatic milking systems reached the market in the 1990s. Jacobs and Siegford (2012) report that they can increase milk production by up to 12%, cut labour by as much as 18% and improve welfare by letting cows choose when to be milked — but farmers may not realise these benefits if some cows do not come to the robot voluntarily or if management routines are not adapted to the system.

Identification in India: Pashu Aadhaar

Autonomous livestock management starts with knowing which animal is which. Under India's **Bharat Pashudhan** digital ecosystem, every registered animal gets a unique 12-digit barcoded ear tag, **Pashu Aadhaar**, used to track breeding, health, ownership and productivity information; 39.23 crore animals had been registered by August 2026 (Press Information Bureau [PIB], 2026). Such records are the labelled data from which future AI tools for Indian herds can learn (Lectures 30–31).

5. Worked example: detecting heat in dairy cows

Missing a cow's heat (oestrus) delays pregnancy and milk. The bovine oestrous cycle lasts 18–24 days (Forde et al., 2011), and cows are much more active during oestrus — about 2.75 to 4 times as active, depending on housing, in a classic pedometer study, with little day-to-day variation in activity when they are not in heat (Kiddy, 1977). An activity monitor can therefore raise an alert when a cow walks far more than her own recent average. The program below simulates 100 days of pedometer data for 20 cows, including weak heats and one day when the whole herd is moved to a new paddock, and tests four alert rules.

Heat (oestrus) detection from activity data, with evaluation, in Python (Lecture 8)

```
import random

random.seed(5)
COWS, DAYS, HERD_EVENT = 20, 100, 50       # day 50: whole herd moved to a new paddock

# 1. Simulate daily activity (pedometer steps) for each cow
activity, heat_days = {}, {}
for cow in range(COWS):
    base = random.uniform(2000, 3000)          # this cow's normal steps per day
    heats, day = set(), random.randint(0, 20)
    while day < DAYS:
        heats.add(day)
        day += random.randint(18, 24)          # oestrous cycle of 18-24 days
    steps = []
    for d in range(DAYS):
        x = base * random.gauss(1.0, 0.10)     # normal day-to-day variation
        if d in heats:
            x *= random.uniform(1.3, 4.0)      # more active in heat (weak to strong)
        if d == HERD_EVENT:
            x *= 1.8                           # everyone walks more on moving day
        steps.append(x)
    activity[cow], heat_days[cow] = steps, heats

# 2. Alert when today's activity is well above the cow's own 7-day average
def evaluate(threshold, use_herd_context):
    tp = fp = fn = 0
    for d in range(7, DAYS):
        ratios = {c: activity[c][d] / (sum(activity[c][d - 7:d]) / 7) for c in range(COWS)}
        herd = sorted(ratios.values())[COWS // 2]           # median cow today
        for c, r in ratios.items():
            if use_herd_context:
                r = r / max(herd, 1.0)                      # discount herd-wide rises
            alert, heat = r >= threshold, d in heat_days[c]
            tp += alert and heat
            fp += alert and not heat
            fn += heat and not alert
    precision, recall = tp / (tp + fp), tp / (tp + fn)
    return tp, fp, fn, precision, recall

print("rule                          TP   FP   FN  precision  recall")
for th in (1.4, 1.8, 2.2):
    tp, fp, fn, p, r = evaluate(th, False)
    print(f"own baseline, ratio >= {th:.1f}   {tp:4d} {fp:4d} {fn:4d}     {p:5.2f}   {r:5.2f}")
tp, fp, fn, p, r = evaluate(1.8, True)
print(f"+ herd context, ratio >= 1.8  {tp:4d} {fp:4d} {fn:4d}     {p:5.2f}   {r:5.2f}")

Output:
rule                          TP   FP   FN  precision  recall
own baseline, ratio >= 1.4     84   19    6      0.82    0.93
own baseline, ratio >= 1.8     74    4   16      0.95    0.82
own baseline, ratio >= 2.2     56    0   34      1.00    0.62
+ herd context, ratio >= 1.8    74    0   16      1.00    0.82
```

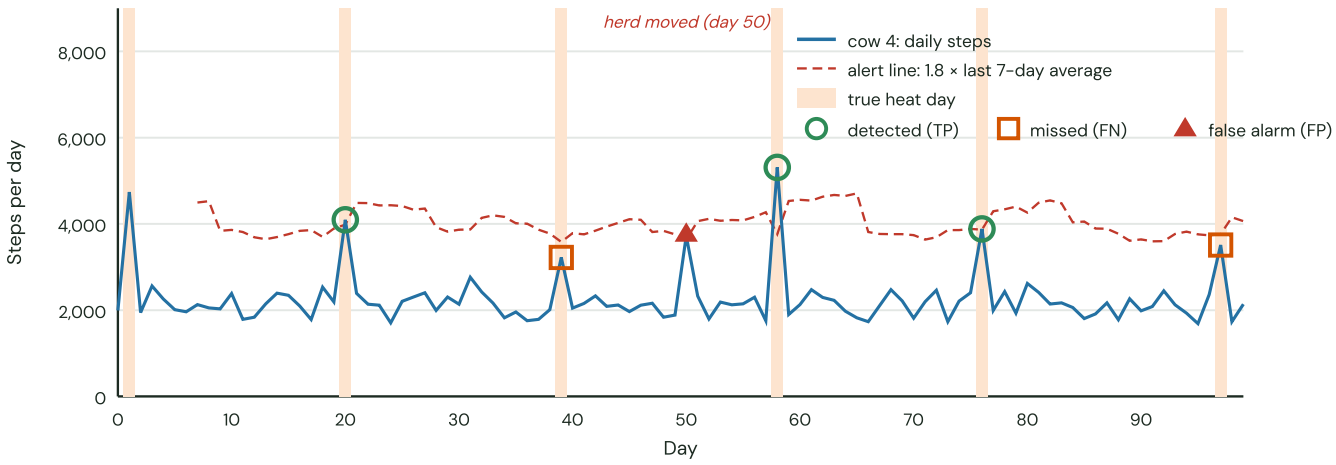


Figure 3. One cow from the program: daily steps, the alert line and the result on each heat day and on the herd-moving day. Original chart generated from the program's data.

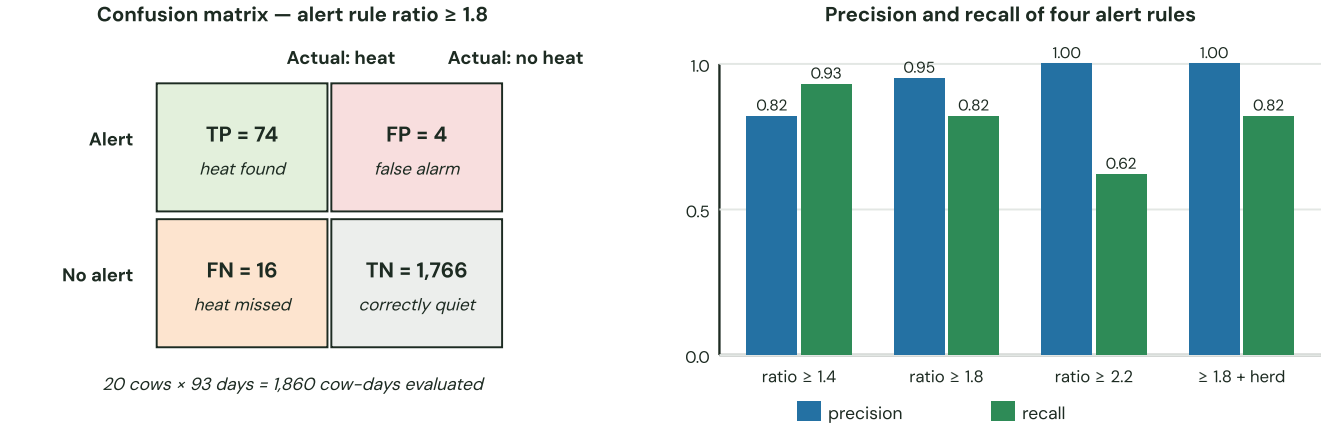


Figure 4. Confusion matrix for the 1.8 rule and the precision–recall trade-off between rules. Original chart from the program's output.

Worked example 2 — judging the 1.8 alert rule
Cow-days evaluated = 20 cows × 93 days (days 7–99, after the first week of baseline) = 1,860.
From the output: TP = 74, FP = 4, FN = 16, so TN = 1,860 - 74 - 4 - 16 = 1,766.
Precision = 74 ÷ 78 = 0.95 — 95% of alerts were real heats. Recall = 74 ÷ 90 = 0.82 — 82% of heats were found.
F1 = 2 × 0.95 × 0.82 ÷ (0.95 + 0.82) = 0.88.
Accuracy = (74 + 1,766) ÷ 1,860 = 98.9% — but a "system" that never raised any alert would still score 1,770 ÷ 1,860 = 95.2%, because heat days are only 4.8% of cow-days. For rare events, judge a model by precision and recall, not accuracy.

- **Lower threshold (1.4)**: more heats found (recall 0.93) but 19 false alarms. **Higher threshold (2.2)**: no false alarms but 34 heats missed. Choosing the threshold is a management decision about which mistake costs more.
- **All 4 false alarms of the 1.8 rule fell on the herd-moving day.** Comparing each cow with the rest of the herd that day removed them without losing any true detections — using more context is how AI systems improve.
- The 16 missed heats were weak ones (activity only 1.2–1.8 times the baseline). Adding other signals — rumination, temperature, milk yield — is how commercial systems try to catch them.

The program was run and the output shown is its actual output. The herd is simulated with illustrative values; real herds, sensors and heats are more variable.

6. Benefits, risks and responsibilities

Benefits	Risks and challenges
Work continues day and night without a driver or milker present	High purchase and repair cost; financial cost is the biggest barrier for small producers (Oliveira et al., 2021)
Precise, consistent operations: less overlap, chemicals placed where needed	Safety of people and animals near machines; need for design standards and supervision (ISO, 2024)
Early warning of heat, illness and stress; better animal welfare (Berckmans, 2014)	Animals or farm routines may not adapt — e.g. cows that do not milk voluntarily (Jacobs & Siegford, 2012)
Less heavy and repetitive work; help where labour is scarce	Effects on rural jobs and labour demand are still little studied (Lowenberg-DeBoer et al., 2020)
Detailed records of every animal and field operation	Models trained on foreign breeds, crops or fields may fail in Indian conditions without local data
Small robots suit small fields and reduce soil load	Dependence on connectivity, positioning signals, power and skilled service

In India, where holdings are small and labour is seasonal, autonomy is most likely to arrive as **services** — robots or drones hired through custom-hiring centres or farmer producer organisations — and as **advisory and monitoring tools** for livestock built on identification systems such as Pashu Aadhaar.

7. Key takeaways

- An autonomous system senses, perceives with AI, decides and acts in a closed loop, under human supervision and with a safety layer.
- Automation ranges from assistance to full autonomy; ISO 18497-1:2024 sets safety design principles for automated farm machinery.
- Machine learning learns patterns from data: supervised, unsupervised and reinforcement learning.
- Models are trained on one set of data and judged on unseen test data; for rare events use precision, recall and F1, not accuracy.
- Field robots combine RTK-GNSS positioning, cameras and LiDAR, path planning, control and safety; most are still at the research stage.
- Small robots working long hours can match a large tractor; economics depend strongly on the farm situation.
- Precision livestock farming monitors animals continuously; milking robots, feeders and activity monitors act or alert automatically.
- Heat detection from activity shows the trade-off between false alarms and missed cases, and the value of extra context.

8. Examination questions

Short-answer questions (2 marks)

Q1. Distinguish automation and autonomy.

Automation repeats programmed actions; autonomy senses, decides and acts by itself to reach a goal in a changing environment.

Q2. Name the four stages of the autonomy loop.

Sense, perceive, decide (plan) and act.

Q3. Name the three main types of machine learning.

Supervised, unsupervised and reinforcement learning.

Q4. Why is a test set kept separate from training data?

To judge the model on cases it has never seen, as it will meet in use.

Q5. Define precision and recall.

Precision = TP/(TP + FP), the share of alerts that are correct; recall = TP/(TP + FN), the share of real cases found.

Q6. What does RTK do in GNSS positioning?

Corrects satellite positions in real time to centimetre-level accuracy.

Q7. What is precision livestock farming?

Continuous, automatic, real-time monitoring and management of animals that warns the farmer when problems arise.

Q8. What is an automatic milking system?

A robotic stall where cows choose when to be milked and a robot attaches the cups and records the milk.

Q9. Why can activity data reveal heat in cows?

Cows are several times more active during oestrus than on other days.

Q10. What is Pashu Aadhaar?

A unique 12-digit barcoded ear tag identifying each registered animal on India's Bharat Pashudhan platform.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the concept of autonomy in agricultural machines with the sense–perceive–decide–act loop and levels of automation.

Definitions; Figure 1; levels table; supervision and safety; ISO 18497-1.

Q2. Explain machine learning, its types and how a model is built and evaluated.

Definition; types table with farm examples; workflow; confusion matrix; precision, recall, F1; accuracy for rare events.

Q3. Describe the components and uses of autonomous field robots and tractors.

Figure 2; localisation, perception, planning, control, communication, safety; review findings; worked example 1.

Q4. Discuss precision livestock farming and automatic milking systems.

PLF aims; systems table; AMS benefits and limits; identification and Pashu Aadhaar.

Q5. Explain how heat in dairy cows can be detected automatically and how such a system is evaluated.

Biology; program logic; Figures 3 and 4; worked example 2; threshold trade-off; herd context.

Q6. Discuss the benefits and risks of autonomous crop and livestock management in India.

Benefits and risks table; economics; safety; local data; services model.

9. Glossary

Term	Meaning
Autonomy	Ability to sense, decide and act towards a goal without continuous human control
Automatic milking system (AMS)	Robotic milking stall that cows visit voluntarily
Confusion matrix	Table of true and false positives and negatives
Deep learning	Machine learning with many-layered neural networks
F1 score	Harmonic mean of precision and recall
Field efficiency	Share of time a machine spends actually working the field
Geofence	Virtual boundary a machine must not cross
IMU	Inertial measurement unit: accelerometers and gyroscopes
LiDAR	Laser sensor that measures distances in 3D
Machine learning	Learning patterns from data instead of hand-written rules
Oestrus (heat)	Period when a cow is ready to mate
Precision	Share of alerts that are correct, TP/(TP + FP)
Precision livestock farming	Real-time automatic monitoring and management of animals
Recall	Share of real cases detected, TP/(TP + FN)
RTK	Real-time kinematic correction for centimetre-level GNSS
Test set	Data kept aside to judge a trained model

References

Berckmans, D. (2014). Precision livestock farming technologies for welfare management in intensive livestock systems. *Revue Scientifique et Technique de l'OIE, 33*(1), 189–196. https://doi.org/10.20506/rst.33.1.2273

Forde, N., Beltman, M. E., Lonergan, P., Diskin, M., Roche, J. F., & Crowe, M. A. (2011). Oestrous cycles in *Bos taurus* cattle. *Animal Reproduction Science, 124*(3–4), 163–169. https://doi.org/10.1016/j.anireprosci.2010.08.025

International Organization for Standardization. (2024). *Agricultural machinery and tractors — Safety of partially automated, semi-autonomous and autonomous machinery — Part 1: Machine design principles and vocabulary* (ISO Standard No. 18497-1:2024). https://www.iso.org/standard/82684.html

Jacobs, J. A., & Siegford, J. M. (2012). Invited review: The impact of automatic milking systems on dairy cow management, behavior, health, and welfare. *Journal of Dairy Science, 95*(5), 2227–2247. https://doi.org/10.3168/jds.2011-4943

Kamilaris, A., & Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture, 147*, 70–90. https://doi.org/10.1016/j.compag.2018.02.016

Kiddy, C. A. (1977). Variation in physical activity as an indication of estrus in dairy cows. *Journal of Dairy Science, 60*(2), 235–243. https://doi.org/10.3168/jds.S0022-0302(77)83859-9

Liakos, K. G., Busato, P., Moshou, D., Pearson, S., & Bochtis, D. (2018). Machine learning in agriculture: A review. *Sensors, 18*(8), 2674. https://doi.org/10.3390/s18082674

Lowenberg-DeBoer, J., Huang, I. Y., Grigoriadis, V., & Blackmore, S. (2020). Economics of robots and automation in field crop production. *Precision Agriculture, 21*(2), 278–299. https://doi.org/10.1007/s11119-019-09667-5

Neethirajan, S. (2017). Recent advances in wearable sensors for animal health management. *Sensing and Bio-Sensing Research, 12*, 15–29. https://doi.org/10.1016/j.sbsr.2016.11.004

Neethirajan, S. (2020). The role of sensors, big data and machine learning in modern animal farming. *Sensing and Bio-Sensing Research, 29*, 100367. https://doi.org/10.1016/j.sbsr.2020.100367

Oliveira, L. F. P., Moreira, A. P., & Silva, M. F. (2021). Advances in agriculture robotics: A state-of-the-art review and challenges ahead. *Robotics, 10*(2), 52. https://doi.org/10.3390/robotics10020052

Press Information Bureau. (2026, August 12). *Digital transformation strengthens livestock and fisheries sectors* [Press release]. Ministry of Fisheries, Animal Husbandry & Dairying, Government of India. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2298199

Russell, S., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson. ISBN 978-0-13-461099-3

All diagrams, examples and programs are original. Machine values in worked example 1 and the simulated herd are illustrative; every calculation and the program output were checked by running them.

Practice MCQs

Q1. An autonomous farm machine differs from a simply automated one because it

(a) repeats a fixed programme
(b) senses, decides and acts by itself towards a goal
(c) needs no sensors
(d) cannot be stopped

Q2. The correct order of the autonomy loop is

(a) act, sense, decide, perceive
(b) sense, perceive, decide, act
(c) decide, act, sense, perceive
(d) perceive, act, sense, decide

Q3. Learning from examples that carry the correct answer is

(a) supervised learning
(b) unsupervised learning
(c) reinforcement learning
(d) rule-based reasoning

Q4. Grouping fields into management zones without labels is an example of

(a) supervised learning
(b) unsupervised learning
(c) reinforcement learning
(d) expert systems

Q5. Precision is calculated as

(a) TP ÷ (TP + FN)
(b) TP ÷ (TP + FP)
(c) (TP + TN) ÷ all cases
(d) FP ÷ (FP + TN)

Q6. In the heat-detection example, why is accuracy (98.9%) misleading?

(a) the program has a bug
(b) heat days are rare, so never alerting would still score 95.2%
(c) precision was zero
(d) recall was 100%

Q7. RTK correction gives GNSS positions accurate to about

(a) a few centimetres
(b) a few metres
(c) a kilometre
(d) 100 m

Q8. In the review by Oliveira et al. (2021), most agricultural robotic systems were

(a) in commercial use
(b) still at the research stage
(c) used for milking
(d) flying drones

Q9. Jacobs and Siegford (2012) report that automatic milking systems can reduce labour by as much as

(a) 5%
(b) 18%
(c) 50%
(d) 90%

Q10. In a classic pedometer study, cows during oestrus were about how many times as active as on other days?

(a) 1.1 times
(b) 2.75 to 4 times
(c) 20 times
(d) no more active

Q11. Pashu Aadhaar is

(a) a 12-digit barcoded ear tag for each registered animal
(b) a milk price app
(c) a cattle breed
(d) a fodder scheme

Q12. ISO 18497-1:2024 deals with

(a) milk quality testing
(b) safety of partially automated, semi-autonomous and autonomous farm machinery
(c) soil testing methods
(d) drone licensing

Answers

Q1 — (b) senses, decides and acts by itself towards a goal
Q2 — (b) sense, perceive, decide, act
Q3 — (a) supervised learning
Q4 — (b) unsupervised learning
Q5 — (b) TP ÷ (TP + FP)
Q6 — (b) heat days are rare, so never alerting would still score 95.2%
Q7 — (a) a few centimetres
Q8 — (b) still at the research stage
Q9 — (b) 18%
Q10 — (b) 2.75 to 4 times
Q11 — (a) a 12-digit barcoded ear tag for each registered animal
Q12 — (b) safety of partially automated, semi-autonomous and autonomous farm machinery